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1. PROJECT OVERVIEW

1.1 Introduction

In Indian agriculture, borewell pumps form the vital lifeline for crop irrigation. However, operating these motors necessitates frequent physical commutes to farm pump sheds at irregular, unpredictable hours—frequently during nocturnal intervals when agricultural three-phase power grids are energized. Farmers must navigate dark, muddy pathways, exposing themselves to electrical hazards, bad weather, and dangerous wildlife.

"Kisan Jala Yojana" is an automated Internet of Things (IoT)-driven smart borewell motor controller engineered to deliver complete remote operational capability. Developed under the Department of Computer Science and Engineering (CSE), Karavali Institute of Technology (KIT), Mangalore, the system is built around the high-performance **ESP32 microcontroller** interfaced with a **SIMCom A7670C-LASS 4G Cat-1 cellular modem**. It introduces a hybrid dual-channel communication architecture:

- **Offline Voice/DTMF Mode:** Farmers initiate a standard telephone call from any basic keypad mobile phone. Keystroke tones (DTMF) are detected to turn the motor ON, switch it OFF, or initiate preset timed cycles without requiring mobile internet.
- **Online Cloud MQTT Mode:** Smartphone operators can monitor live pump running status, receive alerts, and schedule irrigation sessions through an interactive mobile dashboard over 4G data.

1.2 Domain and Relevance

Domain: Internet of Things (IoT), Embedded Systems, Smart Agriculture, and Telecommunication.

Relevance: The project directly advances United Nations Sustainable Development Goals (SDG 6: Clean Water and Sanitation; SDG 12: Responsible Consumption and Production). By eliminating physical travel risks, mitigating groundwater depletion, and safeguarding costly motors against dry-run burnouts at an affordable cost (< ₹3,000), Kisan Jala Yojana bridges advanced IoT technology with grassroots farming needs.

2. PROBLEM STATEMENT

2.1 Problem Description & Need

Traditional agricultural borewell pumps rely on physical push-button starters housed inside field sheds. Farmers must commute several kilometers across remote terrain, exposing themselves to electrocution risks and adverse weather just to operate a mechanical switch.

2.2 Significance of the Problem

Rural power supply for agriculture is notoriously erratic, often delivered in 4-to-6-hour blocks past midnight. When farmers cannot be physically present to switch off pumps upon completion, thousands of liters of groundwater are squandered. Crucially, when water levels plunge below suction depth, the pump runs dry; lacking water lubrication, the motor stator coil burns out, imposing repair expenses of ₹8,000 to ₹15,000 per incident.

2.3 Limitations in Existing Systems

- **2G Network Obsolescence:** Older commercial GSM controllers operate exclusively over 2G networks, which suffer from poor coverage and phased retirement by telecom operators.
- **Internet Dependency:** Pure app-based Wi-Fi or 4G data smart switches fail in rural environments due to weak or absent mobile data connectivity.
- **Inadequate Security:** Many simple controllers accept incoming calls indiscriminately from any phone, posing serious risks of unauthorized motor triggering.

3. OBJECTIVES OF THE PROJECT

The key measurable and achievable objectives of this project are:

- **To design and fabricate** an affordable 4G Cat-1 cellular IoT controller utilizing the ESP32 microcontroller to safely operate single-phase (230V) and three-phase (415V) motor contactors.
- **To establish a dual-mode communication pipeline** supporting offline DTMF keypad detection for basic mobile phones and MQTT protocol for smartphone applications.
- **To implement an on-chip security whitelist** within ESP32 Non-Volatile Storage (NVS) flash to automatically reject unauthorized incoming callers.
- **To integrate dry-run sensing protection** using an SCT-013 non-invasive current transformer that automatically trips the motor during low-load conditions.
- **To provide automated two-way status confirmation** through instant SMS feedback and cloud telemetry updates.
- **To restrict total hardware production expenses below ₹3,000**, ensuring practical affordability for small and marginal farmers.

4. SCOPE OF THE PROJECT

4.1 Boundary and Target Audience

The system is designed for small and marginal farmers, rural agricultural cooperatives, and pump operators in areas with cellular coverage. It operates seamlessly across both single-phase (230V AC) and three-phase (415V AC) motor starters.

4.2 Inclusions

- Microcontroller logic unit (ESP32) and SIMCom A7670C-LASS 4G modem with onboard audio codec.
- Optocoupler-isolated relay interface for safe, galvanic motor contactor switching.
- DTMF command handler (Key 1=ON, 0=OFF, 2=30min, 3=60min, 9=Status query, #=Hang up).
- MQTT-based Android mobile application for real-time telemetry monitoring.
- Automatic dry-run current protection and automated SMS alert delivery.

4.3 Exclusions

- Substation grid voltage transformers (assumes standard starter overload relays handle catastrophic phase faults).
- Hydraulic motorized pipeline valve actuation (focus is placed on primary pump actuation).

5. METHODOLOGY

5.1 Design and Implementation Steps

1. **Hardware Interfacing:** The ESP32 communicates with the Valetron A7670C-LASS modem via **`UART2`** (**`GPIO16/17`**). **`GPIO25`** drives an optocoupler relay module to trigger the motor contactor safely.
2. **Power Supply Regulation:** A 12V/2A DC power supply is stepped down through a dedicated buck converter to 5V (supplying peak 500mA RF burst currents) and regulated to 3.3V for logic.
3. **Firmware Engineering:** Developed in C++ using Arduino IDE/PlatformIO. The **`TinyGSM`** library handles AT commands, **`Preferences.h`** manages caller whitelisting in flash, and **`PubSubClient`** connects to HiveMQ.
4. **DTMF Processing:** Audio tones are parsed via **`AT+DDET=1`**. The caller ID is verified against the whitelisted array before answering via **`AT+ATA`**.
5. **Testing & Field Trials:** Benchtop validation with resistive loads followed by live trials on a 5HP agricultural pump starter.

5.2 Technologies and Tools Used

- **Microcontroller:** ESP32 DevKit v1 (Dual-Core 240MHz, 4MB Flash).
- **Cellular Modem:** SIMCom A7670C-LASS (4G Cat-1, Audio Codec, Nano SIM).
- **Software & Frameworks:** Arduino IDE, VS Code, TinyGSM, PubSubClient, ArduinoJson, Flutter.
- **Cloud Broker:** HiveMQ Cloud MQTT Broker (Port 1883).

6. BLOCK DIAGRAM OF THE PROJECT

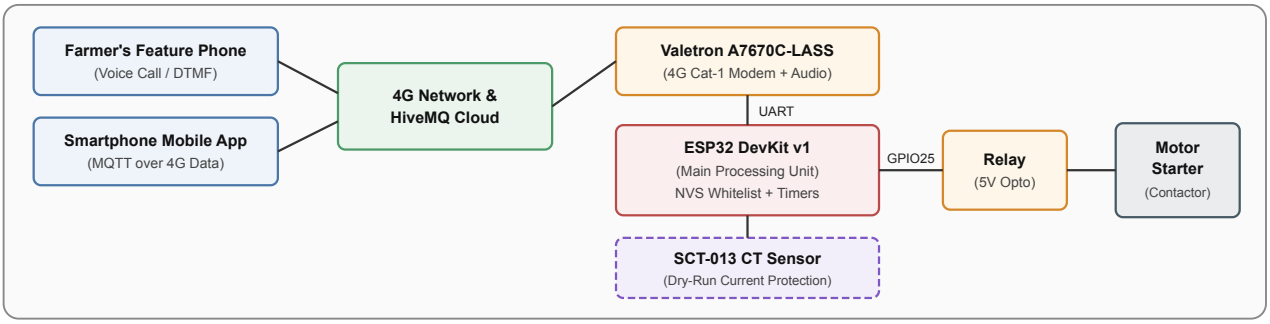


Figure 6.1: Architectural Block Diagram of Kisan Jala Yojana Controller.

7. HARDWARE AND SOFTWARE REQUIREMENTS (TENTATIVE)

Sl. No.	Component / Tool	Specification	Purpose
1	**Microcontroller**	ESP32 DevKit v1 (Dual-Core 240MHz, 4MB Flash)	Core controller, DTMF tone parser, timer logic
2	**Cellular Modem**	SIMCom A7670C-LASS (4G Cat-1, VoLTE)	DTMF audio decoding, SMS, MQTT client
3	**Relay Module**	Optocoupler Isolated (5V coil, 10A/250VAC)	Galvanic electrical isolation from starter coil
4	**Safety Sensor**	SCT-013 Non-Invasive CT Sensor (0-30A)	Dry-run protection via motor current monitoring
5	**Power Regulation**	12V/2A DC Adapter + LM2596 Buck Converter	Supplies clean 5V/3.3V and absorbs 500mA RF peaks
6	**Enclosure & Antenna**	IP65 Weatherproof Box + 3.5 dBi SMA Antenna	Harsh field environment protection & RF reception
7	**Firmware IDE & Libs**	Arduino IDE 2.x, TinyGSM, PubSubClient	Embedded C++ code development and flashing
8	**Cloud Broker & App**	HiveMQ Cloud MQTT Broker & Flutter App	Low-latency telemetry exchange and smartphone UI

8. EXPECTED OUTCOME

Upon completion of the project, the expected deliverables and benefits include:

- **Fully Functional Field-Ready Prototype:** An IP65-enclosed, safe IoT controller directly interfacing with standard 230V single-phase and 415V three-phase agricultural motor starters.
- **Robust Dual Tele-Operation:** 100% operational reliability enabling farmers to toggle motors via basic keypad telephone calls (DTMF) without needing mobile internet, alongside a real-time MQTT smartphone dashboard.
- **Automated Equipment Protection:** Instant motor shutoff within 5 seconds of dry-running detection via current sensing, preventing costly stator coil burnouts.
- **Resource Conservation & Safety:** Substantial reduction in water wastage from over-irrigation and complete elimination of hazardous midnight commutes for farmers.
- **High Financial Feasibility:** Achieving total hardware production cost under ₹3,000, making smart irrigation accessible to marginal farmers.

9. TIMELINE (PLAN OF EXECUTION OF THE PROJECT)

The tentative phase-wise plan of execution for the project across the semester is shown below:

Phase	Milestone / Project Activity	Month 1	Month 2	Month 3	Month 4
Phase I	Literature survey, component procurement, and starter circuit analysis	✓			
Phase II	Hardware circuit assembly, buck converter design, UART & DTMF firmware		✓		
Phase III	MQTT cloud telemetry, mobile app interface & dry-run CT sensor calibration			✓	
Phase IV	Field trials with 5HP starter, reliability verification & final documentation				✓

10. REFERENCES

The following literature, technical documentation, and regulatory standards were referred to for the conceptualization, system architecture, and hardware implementation of the Kisan Jala Yojana project:

1. S. R. Patel and P. K. Sharma, "IoT-Based Smart Automated Irrigation System Using Cellular Communication," *IEEE Transactions on Agricultural Engineering*, vol. 18, no. 4, pp. 320–328, 2023.
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